

Houghton's group H_3 has superpolynomial sofic profile

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Abstract

Amenable groups can be approximated by permutations, but the approximations need not be small. The sofic profile of a finite piece of a group measures how large a permutation model must be to reproduce its multiplication up to an error $1/r$. We show that Houghton's group H_3 , a finitely presented elementary amenable group, has a finite piece whose sofic profile grows at least as fast as $\exp(r^c)$ with $c = 1/(\log_2 130 + 1) \approx 1/8.02$, and at most like $\exp(O(\sqrt{r} \log r))$. Since H_3 is amenable, and hence sofic, this answers the last question of Cornulier's Problem 3.18: a sofic group can have superpolynomial sofic profile. It also shows that Cornulier's extension theorem, from which he derived that every elementary amenable group has polynomial sofic profile, is false as stated, and we locate the error: the finite piece of the kernel that it provides must grow with the error parameter, and we give the corrected, scale-dependent form of the bound. The same argument bounds unitary approximations, where our unitary models, which spread a defect thinly, give a better upper bound than our permutation construction. The upper sofic bound holds for every fixed finite chunk of H_3 . The proof combines a polynomial bound on the area of far commutators in Johnson's presentation, obtained from a doubling self-similarity with executable certificates for its finite relations, with an entropy count for approximately commuting copies of the symmetric group S_3 . A collection argument also gives the full Dehn-function bound $\delta_{H_3}(n) = O(n^{\log_2 130 + 4})$.

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1 Introduction

Sofic groups are the groups whose finite pieces can be modelled by permutations up to a small error, and hyperlinear groups those whose finite pieces can be modelled by unitary matrices. Quantitative versions ask, for groups that can be approximated, how large the models must be. Cornulier introduced the *sofic profile* $\sigma_E(r)$ of a finite piece (a chunk) E of a group: the least n for which E maps to the symmetric group Sym_n with multiplication defect $1/r$ and with distinct elements at normalized Hamming distance at least $1 - 1/r$ [2]. Amenable groups are sofic. More quantitatively, Cornulier derived from an extension theorem [2, Theorem 3.15] that every elementary amenable group has polynomial sofic profile [2, Example 3.16], and asked for methods to bound the sofic profile of explicit groups from below, whether some group has sofic profile that is unbounded and not linear in r , and whether such a group can be sofic [2, Problem 3.18].

Houghton's group H_3 is finitely presented and elementary amenable: it is an extension of the locally finite group $\text{FSym}(\mathbb{N})$ of finitary permutations by $\mathbb{Z}^2$ [5]. Section 2 defines it and a chunk E_H .

Theorem 1.1. *The chunk E_H of H_3 has sofic profile*

$$2^{r^{1/(\log 130+1)}/940-1/16} \leq \sigma_{E_H}(r) \leq 2^{O(\sqrt{r} \log r)} \quad (r > 1).$$

The unweighted bound obtained from certified finite fillings already gives $\sigma_{E_H}(r) \geq 2^{r^{1/12}/270-1/16}$.

Here and below logarithms are to base two, and $\log 130 \approx 7.022$.

Corollary 1.2. *Theorem 3.15 of [2] is false for the extension $1 \rightarrow \text{FSym}(\mathbb{N}) \rightarrow H_3 \rightarrow \mathbb{Z}^2 \rightarrow 1$ and the chunk E_H . So H_3 is an elementary amenable group whose sofic profile is not polynomial, contrary to [2, Example 3.16].*

We locate the error in Section 7: the extension theorem asserts a finite piece of the kernel that does not depend on the error parameter, but the construction it relies on needs kernel elements conjugated along a Følner set whose size grows with the inverse error. Tracking that dependence gives a correct, scale-dependent extension bound (Proposition 7.2). Since H_3 is amenable, and hence sofic, Theorem 1.1 also answers the last question of Cornulier's Problem 3.18: a sofic group can have sofic profile that is unbounded and not linear in r . Slofstra constructed the first finitely presented group with superpolynomial hyperlinear and sofic profile [7], answering the first question; that group is not amenable, and its soficity and even hyperlinearity were left unresolved there. To our knowledge H_3 is the first amenable group shown to have superpolynomial sofic profile.

The lower bound holds for unitary approximations as well. Write $P = \langle a, b, \alpha \mid r_1, \dots, r_5 \rangle$ for Johnson's presentation (Section 2) and call unitaries $\phi(a), \phi(b), \phi(\alpha) \in U(D)$ an *e-representation* if every relator satisfies $\|\phi(r_i) - I\|_2 \leq e$ in the normalized Hilbert–Schmidt norm.

Theorem 1.3. *Let $\beta_H = 1/(2 \log 130 + 1) \approx 1/15.04$, $K' = 750 \cdot 30 \cdot 3^{\log 130}$ and $e > 0$. Every e-representation of P on $\mathbb{C}^D$ in which the three-cycle $\gamma = \alpha a \alpha a^{-1}$ satisfies $\|\phi(\gamma) - I\|_2 \geq 1/2$ has $\log D \geq (K'e)^{-2\beta_H}/16 - 1/16$.*

Conversely, spreading the single defect of our permutation models over many cells gives such representations with $\log D = O(e^{-2/3})$, while our permutation models give only $O(e^{-1})$ (Corollary 6.6).

How the proofs go.

- *Far commutators have polynomial area* (Section 3). The commutator of two conjugates of the basic transposition at distance R has area at most $30R^{\log 130}$ in Johnson’s presentation, by a doubling self-similarity of H_3 whose finite relations are certified by machine. Buffered collection upgrades these estimates to a polynomial bound on the full Dehn function (Theorem 3.3), improving the exponential upper bound of [4].
- *Displaced copies of S_3 need dimension* (Section 4). An approximate representation gives, through the far commutators, many approximately commuting copies of S_3 whose three-cycles are displaced from the identity. Rounding each copy to an exact one and counting the entropy needed to make a state invariant forces dimension exponential in their number.
- *Transport* (Section 5). For permutations, relator errors add up in measure over a van Kampen diagram; for unitaries, in norm. This converts the area bound into the two exponents.
- *Upper bounds* (Section 6). Permutation models on a torus of clock positions that fail at a single corner, and unitary models that spread that failure over many cells by small rotations.

Related work. Quantitative metric approximations of groups are surveyed in [9]; the hyperlinear profile and its relation to nonlocal games are studied in [8, 7], and hyperlinear approximations of amenable groups come from sofic ones [10]. Our method follows Slofstra’s in combining stability of approximate representations of finite groups, here an entropy bound in place of the stability theorems of Gowers–Hatami [11] and De Chiffre–Ozawa–Thom [12], with an isoperimetric input. Houghton’s groups have an exponential isoperimetric inequality [4] and are not co-Hopfian [6]; we use the doubling endomorphism behind the latter. The extension theorem is attributed in [2] to the construction of Elek and Szabó [3]. A companion paper [14] applies the lower bounds to the memory needed for the repeated use of a quantum channel built from H_3 .

Organization. Section 2 defines H_3 , the chunk and the profiles. Section 3 proves the area bounds, Section 4 the dimension bounds for approximate representations, and Section 5 the lower bounds of Theorems 1.1 and 1.3. Section 6 gives the upper bounds, Section 7 the correction, and Section 8 open problems. Appendix A gives the fillings.

2 Houghton’s group, chunks and profiles

Let $Y = \{(i, j) : i \in \{1, 2, 3\}, j \geq 1\}$ be three rays. Permutations act on the right and gh means g followed by h . Let a send $(1, 1) \mapsto (2, 1)$, $(1, j) \mapsto (1, j - 1)$ for $j \geq 2$ and $(2, j) \mapsto (2, j + 1)$, fixing ray 3; let b be defined in the same way with rays 2 and 3 exchanged; let α be the transposition of $(1, 1)$ and $(1, 2)$. These generate H_3 , the group of permutations of Y that are eventually translations on each ray. They satisfy

$$r_1 = \alpha^2, \quad r_2 = (\alpha a \alpha a^{-1})^3, \quad r_3 = [\alpha, a^2 \alpha a^{-2}], \quad r_4 = a b a^{-1} b^{-1} \alpha^{-1}, \quad r_5 = a \alpha a^{-1} b \alpha^{-1} b^{-1}, \quad (1)$$

with $[x, y] = xyx^{-1}y^{-1}$; each identity can be checked directly on the finitely many points where the word moves anything. By a theorem of Johnson (see [4, Theorem 2.8]) these relators present H_3 . The lower bound below uses only that $r_1, \dots, r_5$ hold in H_3 , the far-commutator areas of Section 3, and that the three-cycle $\gamma = \alpha(a\alpha a^{-1})$ on $(1, 1), (1, 2), (1, 3)$ is nontrivial. The short exact sequence $1 \rightarrow \text{FSym}(\mathbb{N}) \rightarrow H_3 \rightarrow \mathbb{Z}^2 \rightarrow 1$ shows that H_3 is elementary amenable.

We recall Cornulier's definitions [2, Definitions 3.2 and 3.4]. A *chunk* of a group G is a finite subset $E \ni 1$ with the partial multiplication $xy = z$ whenever $x, y, z \in E$ and $xy = z$ in G . For a map $f : E \rightarrow \text{Sym}_n$ with $f(1) = \text{id}$, let d_n be the normalized Hamming distance on Sym_n . The map is an ε -*morphism* if $d_n(f(xy), f(x)f(y)) \leq \varepsilon$ whenever xy is defined in E , and it is $(1 - \varepsilon)$ -*expansive* if $d_n(f(x), f(y)) \geq 1 - \varepsilon$ for distinct $x, y \in E$. The sofic profile is

$$\sigma_E(r) = \min\{n : \text{there is a } (1 - r^{-1})\text{-expansive } r^{-1}\text{-morphism } E \rightarrow \text{Sym}_n\}, \quad r > 1.$$

A group has polynomial sofic profile if every chunk has $\sigma_E \preceq$ some polynomial [2, Definition 3.10].

Let $E_H \subset H_3$ be the symmetric chunk consisting of $1, a^{\pm 1}, b^{\pm 1}, \alpha^{\pm 1}$, all prefixes of the relators $r_1, \dots, r_5$ and of the word $\alpha a \alpha a^{-1}$ for γ , and the inverses of these elements.

For permutations, relations can be transported in measure: the normalized Hamming distance d_n is bi-invariant, so a null word that is a product of A conjugates of relators has $d_n(\phi(w), \text{id}) \leq A\delta$ when every relator has $d_n(\phi(r_i), \text{id}) \leq \delta$. For unitaries only $\|\phi(w) - I\|_2 \leq A \max_i \|\phi(r_i) - I\|_2$ is available in general, since $\|(e^{i\theta} I)^A - I\|_2$ grows like $A\theta$.

3 Polynomial area of far commutators

Let $P = \langle a, b, \alpha \mid r_1, \dots, r_5 \rangle$, put $\sigma_j = a^j \alpha a^{-j}$, and let $A^*(M)$ be the largest area in P of a commutator $[\sigma_p^{\pm 1}, \sigma_q^{\pm 1}]$ with $0 \leq p, q \leq M$ and $|p - q| \geq 2$. These commutators are trivial in H_3 because the transpositions have disjoint supports.

Theorem 3.1 (Polynomial far-commutator area). $A^*(M) \leq M^{11}$ for every $M \geq 2$.

Outline; the full proof is in Appendix A. Let ψ be the endomorphism of the free group on a, b, α with $\psi(a) = a^2$, $\psi(b) = b^2$ and $\psi(\alpha) = [a^2, b^2]$. On H_3 it is induced by the diagonal action on two copies of Y , identified with Y by the bijection $(1, j, e) \mapsto (1, 2j - 1 + e)$ and $(i, j, e) \mapsto (i, 2j - e)$ for $i = 2, 3$ and $e \in \{0, 1\}$. The resulting group endomorphism, not this bijection of sets, is injective and not surjective (compare [6]). The proof has four steps.

- (1) *ψ lifts to P with bounded area multiplier.* Each $\psi(r_i)$ is a null word in P , of area at most 0, 17, 182, 864, 359 for $i = 4, 1, 5, 3, 2$. Hence $\text{Area}_P(\psi(w)) \leq 864 \text{Area}_P(w)$.
- (2) *Two cheap letters.* With $t = b^{-1}a$ and $\rho = a^{-1}b^{-1}ab$, the letter α commutes with t at cost 1 and with ρ at cost at most 5, and $\{t, \rho\}$ is invariant under ψ up to one relation, $b^{-1}\rho b = t\rho t^{-1} \cdot \rho \cdot t\rho t^{-1}$, of area at most 14. The identities $\psi(t) = t\rho t$, $t\rho = b^{-1}tb$ and $\psi(\rho) = t^{-1}\rho^b \rho^{b^2} t \rho \rho^b$ hold in the free group.
- (3) *A doubling recursion.* The words $X_n = b^{-n}a^n$ satisfy $X_{2n} = \psi(X_n)$ and $X_{2n+1} = b^{-1}X_{2n}bt$ freely, and in H_3 act as a reversal of $[1, n]$ followed by translation on the line formed by rays 2 and 3. Writing X_n as a word W_n in t, ρ costs $R(n)$ cells, with $R(2n) \leq 864 R(n) + 84 \#_\rho(W_n)$ and a comparable odd step. Unrolling gives $R(n) \leq 1.35 n^{\log_2 864}$.
- (4) *From X_n to far commutators.* $[\alpha, X_n]$ expands into one conjugate of $[\alpha, t]^{\pm 1}$ or $[\alpha, \rho]^{\pm 1}$ per letter of W_n , so its area is at most $2R(n) + 5|W_n|$; a far commutator $[\sigma_0, \sigma_m]$ reduces

to it at linear extra cost. This gives $A^*(m) \leq 5.4 m^{\log_2 864} + 10 m^{\log_2 323} + 18m \leq m^{11}$ for $m \geq 5$, and $A^*(2), A^*(3), A^*(4) \leq 1, 21, 47$.

□

Certificates. Every finite relation used in the proof is supplied in machine-checkable form: as an explicit list of cells (z_k, r_{i_k}, s_k) such that the relation equals $\prod_k z_k r_{i_k}^{s_k} z_k^{-1}$ in the free group on a, b, α , which a short independent program verifies by free reduction after recomputing the relation from its definition. The certified relations and their cell counts are: $A(2) = 1$, $A(3) \leq 21$, $A(4) \leq 47$, $A(5) \leq 79$, $A(6) \leq 157$; $E(0..3) \leq 1, 4, 27, 76$; $U(1..3) \leq 1, 5, 32$; $B(2), B(3) \leq 9, 12$; the relation of step (2) with 14 cells; $[\alpha, \rho]$ with 5; $\psi(\alpha) = \sigma_1 \sigma_2 \sigma_0 \sigma_1$ with 6; the images $\psi(r_1), \psi(r_2), \psi(r_3), \psi(r_5)$ with 17, 359, 864, 182; and the rewriting of $\psi(\rho)$ as a word in t and ρ , which uses the relation of step (2) six times, with 84. The recursion of step (3) has also been executed mechanically, transporting certificates through ψ cell by cell: for $n \leq 9$ the certificates of $X_n = W_n$ have 0, 0, 14, 84, 392, 995, 2493, 5298, 11878 cells, each at most the recursion bound, and the resulting far commutators satisfy $A(m) \leq 73\,121$ for $m \leq 11$, far below m^{11} . Appendix A gives the fillings in words; the ancillary files contain the generator, the verifier and the exported certificates. The exported file holds every finite relation listed above and the recursion for $n \leq 5$, which the independent verifier checks; the instances $6 \leq n \leq 9$ were executed by the generator but are not in the file, and the generator exports them for the verifier on request. The proof uses only the finite relations and the hand induction of Appendix A.

The same fillings give a better exponent when the five kinds of cells are counted separately.

Theorem 3.2 (Reweighted far-commutator area). $A^*(M) \leq 30 M^{\log 130}$ for every $M \geq 2$.

The proof is in Appendix A. The same area estimates also control arbitrary null words.

Theorem 3.3 (Polynomial Dehn function). *Let $\delta_P(n)$ be the maximum area of a null word of length at most n in Johnson's presentation. There is an absolute constant C_0 such that*

$$\delta_P(n) \leq C_0 n^4 (1 + A^*(4n + 2)) = O(n^{\log 130 + 4}) \quad (n \geq 1).$$

The unweighted estimate of Theorem 3.1 alone gives $\delta_P(n) = O(n^{15})$.

The proof in Appendix B collects a buffered null word into adjacent transpositions, then fills it using a uniform Coxeter insertion algorithm. This full Dehn bound does not improve the profile exponents: the direct far-commutator estimate is sharper for the words used there. Reweighting the far-commutator fillings cannot bring their exponent below $\log \rho \approx 7.018$, where $\rho \approx 129.64$ is the spectral radius of their substitution matrix; $\log 130 \approx 7.022$ is the certified choice. The area is at least quadratic.

Proposition 3.4 (Quadratic lower bound). *For odd $k \geq 9$, $\text{Area}(C_k) \geq (k - 7)(k - 5)/8$, where $C_k = [\alpha, a^k \alpha a^{-k}]$. Hence $A^*(R) \geq R^2/16$ for every integer $R \geq 23$.*

Proof. In a permutation model in which every relator moves at most a fraction δ of the points, a null word of area A moves at most a fraction $A\delta$ (transport in measure, Section 2). In the corner models of Proposition 6.1, $\delta \leq M^{-2}$ while C_{2M-1} moves a fraction at least $(1 - s^{-2})(M - 4)(M - 3)/(2M^2)$. Hence $\text{Area}(C_{2M-1}) \geq (1 - s^{-2})(M - 4)(M - 3)/2$ for every s , which gives the claim with $k = 2M - 1$. For integer $R \geq 23$ apply it to the largest odd $k \leq R$. □

Thus the present uniform-area transport and sector estimate cannot give more than the exponent $1/3$ in r for permutation approximations, and $2/5$ in $1/e$ for unitary approximations with relator defect e (Section 8). This is a limit of that argument, not of every possible use of far commutators.

4 Stability and counting

The full S_3^N sector bound below feeds the permutation-profile estimate; the phase-robust version later in the section feeds the unitary estimate and the companion's weighted extraction. The entropy arguments use monotonicity of the Holevo quantity under partial trace, a consequence of strong subadditivity [1].

4.1 An entropic sector bound

Lemma 4.1 (Projection products). *Let $P_1, \dots, P_m$ be orthogonal projections on a Hilbert space and $\|x\| = 1$. Then $1 - \operatorname{Re}\langle x, P_m \cdots P_1 x \rangle \leq 2 \sum_j \|(I - P_j)x\|^2$.*

Proof. Put $x_0 = x$, $x_j = P_j x_{j-1}$ and $r_j = x_{j-1} - x_j \in \operatorname{ran}(I - P_j)$. Then $\sum_j \|r_j\|^2 = 1 - \|x_m\|^2 =: b$, and $a = 1 - \operatorname{Re}\langle x, x_m \rangle = \operatorname{Re} \sum_j \langle (I - P_j)x, r_j \rangle \leq \sqrt{Eb}$ with $E = \sum_j \|(I - P_j)x\|^2$. Since $0 \leq \|x - x_m\|^2 = 2a - b$, we get $a \leq \sqrt{2Eb}$, that is $a \leq 2E$. $\square$

Lemma 4.2 (Entropic sector bound). *Let $u_1, \dots, u_N : S_3 \rightarrow U(D)$ be unitary representations, not assumed to commute, such that for a three-cycle c*

$$\|u_i(c) - I\|_2 \geq \Delta \quad (1 \leq i \leq N), \quad \|u_i(g)u_j(h) - u_j(h)u_i(g)\|_2 \leq \eta \quad (i \neq j, g, h \in S_3).$$

Put $t = \sqrt{2N} \eta$. If $t \leq \Delta$ and $t^2 \leq 5/6$, then

$$\log D \geq N \left[\frac{(\Delta - t)^2}{3} - h_2(t^2) - t^2 \log 5 \right].$$

Proof. Let $G = S_3^N$ and $f(g) = u_1(g_1) \cdots u_N(g_N)$. For h in factor i , $\|f(gh) - f(g)u_i(h)\|_2 = \|S^*XS - X\|_2$ with $X = u_i(h)$ and $S = u_{i+1}(g_{i+1}) \cdots u_N(g_N)$, and $\|S^*XS - X\|_2^2 = 2 - 2 \operatorname{Re} \tau(X^*S^*XS)$. The averages $T_j(Y) = \frac{1}{6} \sum_a u_j(a)^* Y u_j(a)$ are orthogonal projections on M_D with the normalized Hilbert–Schmidt inner product, the average of S^*XS over g is $T_N \cdots T_{i+1}(X)$, and $\|(I - T_j)X\|_2^2 = \frac{1}{12} \sum_a \|X, u_j(a)\|_2^2 \leq \eta^2/2$. Lemma 4.1 therefore gives

$$\mathbb{E}_g \|f(gh) - f(g)u_i(h)\|_2^2 \leq 2(N - i)\eta^2 \leq t^2.$$

Let R be the right regular representation of G and $W : \mathbb{C}^D \rightarrow \ell^2(G) \otimes \mathbb{C}^D$ the isometry $Wv = |G|^{-1/2} \sum_g |g\rangle \otimes f(g)v$. Then $\|R(h)W - Wu_i(h)\|_{HS} \leq \sqrt{D}t$ for h in factor i . Put $P = WW^*$ and $\rho = P/D$, a state of entropy $\log D$. For the three-cycle c_i of factor i and $Z_i = 2I - R(c_i) - R(c_i)^*$,

$$\operatorname{Tr}(\rho Z_i) = D^{-1} \|(R(c_i) - I)W\|_{HS}^2 \geq (\Delta - t)^2.$$

Let $\mathcal{E}_i(X) = \frac{1}{6} \sum_h R(h_i)XR(h_i)^*$, with h_i the element h placed in factor i ; these conditional expectations commute. Since $(I - P)W = 0$, $\operatorname{Tr}[(I - P)\mathcal{E}_i(\rho)] \leq t^2$. The state $\mathcal{E}_i(\rho)$ is an average of six states of rank D , one of them ρ , so after pinching by P and $I - P$ its two blocks have rank at most D and $5D$. Pinching does not decrease entropy and $x \mapsto h_2(x) + x \log 5$ increases on $[0, 5/6]$, so $S(\mathcal{E}_i(\rho)) \leq \log D + h_2(t^2) + t^2 \log 5$. For a conditional expectation $S(\mathcal{E}(\sigma)) - S(\sigma) = D(\sigma \|\mathcal{E}(\sigma))$; since the $\mathcal{E}_j$ commute, monotonicity of relative entropy under $\mathcal{E}_{i-1} \cdots \mathcal{E}_1$ bounds the entropy increment of $\mathcal{E}_i$ applied after $\mathcal{E}_{i-1} \cdots \mathcal{E}_1$ by $D(\rho \|\mathcal{E}_i \rho)$. Hence $\bar{\rho} = \mathcal{E}_N \cdots \mathcal{E}_1(\rho)$ satisfies $S(\bar{\rho}) \leq \log D + N[h_2(t^2) + t^2 \log 5]$.

The state $\bar{\rho}$ commutes with $R(G)$, so $\bar{\rho} = \bigoplus_\lambda p_\lambda (I_{d_\lambda}/d_\lambda \otimes \sigma_\lambda)$ over the irreducible representations λ of S_3^N , and $S(\bar{\rho}) \geq \sum_\lambda p_\lambda \log d_\lambda = \sum_\lambda p_\lambda k_\lambda$, where k_λ is the number of factors of λ carrying the two-dimensional representation of S_3 . The operator Z_i is central in the group algebra, so $\operatorname{Tr}(\bar{\rho} Z_i) = \operatorname{Tr}(\rho Z_i)$, and it acts as $3I$ on the sectors whose factor i is two-dimensional and as 0 on the others. Hence $N(\Delta - t)^2 \leq \sum_i \operatorname{Tr}(\bar{\rho} Z_i) = 3 \sum_\lambda p_\lambda k_\lambda \leq 3S(\bar{\rho})$. $\square$

Lemma 4.3 (Exact S_3 rounding). *If $A, B \in U(D)$ satisfy $\|A^2 - I\|_2, \|B^2 - I\|_2, \|(AB)^3 - I\|_2 \leq e$, there is a unitary representation u of $S_3 = \langle s, t \mid s^2, t^2, (st)^3 \rangle$ on $\mathbb{C}^D$ with $\|u(s) - A\|_2 \leq e$, $\|u(t) - B\|_2 \leq 8e$ and $\|u(st) - AB\|_2 \leq 9e$, and each normal form $1, s, t, st, ts, sts$ evaluated in u is within $10e$ of the same word in A, B .*

Proof. Round the spectra of A and B to the nearer of ± 1 . For $|z| = 1$ this moves z by at most $|z^2 - 1|$, so the resulting Hermitian unitaries S, T satisfy $\|S - A\|_2, \|T - B\|_2 \leq e$. Then $C = ST$ satisfies $SCS = C^*$ and $\|C^3 - I\|_2 \leq 7e$. Round the spectrum of C to the nearest cube root of unity, choosing 1 at ties and at -1 . This rule commutes with complex conjugation and moves z by at most $|z^3 - 1|$, so C_0 satisfies $C_0^3 = I$, $SC_0S = C_0^*$ and $\|C_0 - C\|_2 \leq 7e$. Then $u(s) = S$ and $u(t) = SC_0$ are involutions with $u(s)u(t) = C_0$, and the bounds follow from the triangle inequality. $\square$

4.2 A phase-robust form

The count of Lemma 4.2 can be run on products of three-cycles alone. It then needs no relation between the transpositions of different copies, so a coherent phase in their commutators does no harm. Write $\kappa(u, v) = \|uv - vu\|_2$ and $G(t) = h_2(\min\{\frac{2}{3}t, \frac{2}{3}\}) + \min\{\frac{2}{3}t, \frac{2}{3}\}$.

Theorem 4.4 (Phase-robust sector bound). *Let $c_i, x_i \in U(D)$, $1 \leq i \leq N$, satisfy $c_i^3 = x_i^2 = I$ and $x_i c_i x_i = c_i^{-1}$, with no condition on the commutators of x_i and x_j . Put $\Delta_i = \|c_i - I\|_2$ and $f(g) = c_1^{g_1} \cdots c_N^{g_N}$ for $g \in \mathbb{Z}_3^N$, let $g^{(i)}$ be g with its i -th coordinate negated, and, for uniform g ,*

$$\hat{a}_i^2 = \mathbb{E}_g \|f(g + e_i) - f(g)c_i\|_2^2, \quad \hat{b}_i^2 = \mathbb{E}_g \|x_i f(g^{(i)}) - f(g)x_i\|_2^2.$$

Then $\hat{a}_i^2 \leq \frac{4}{3} \sum_{j>i} \kappa(c_i, c_j)^2$, $\hat{b}_i^2 \leq \frac{4}{3} \sum_{j \neq i} \kappa(x_i, c_j)^2$, and

$$\log D \geq \sum_{i=1}^N \left[\frac{(\Delta_i^2 - \hat{a}_i^2)_+}{3} - \frac{\hat{b}_i^2}{2 \ln 2} - G(\hat{a}_i^2) \right].$$

In particular $\Delta_i \geq 49/100$ and $\hat{a}_i, \hat{b}_i \leq 1/100$ for all i give $\log D \geq N/16$.

Proof. *The ordered-product errors.* For h unitary, $\|(I - E_j)h\|_2^2 = \frac{1}{3} \kappa(h, c_j)^2$ for the average E_j of conjugation by the powers of c_j . Moving c_i through the suffix $c_{i+1}^{g_{i+1}} \cdots c_N^{g_N}$ and averaging over its exponents gives $\hat{a}_i^2 = 2 - 2 \operatorname{Re} \langle c_i, E_N \cdots E_{i+1} c_i \rangle$, and Lemma 4.1 gives the first bound. For the second, $T_j(Z) = \frac{1}{3} \sum_k c_j^{-k} Z c_j^{s_j k}$ with $s_i = -1$ and $s_j = 1$ otherwise are orthogonal projections, $T_i(x_i) = x_i$ by the local relation, $\|(I - T_j)x_i\|_2^2 = \frac{1}{3} \kappa(x_i, c_j)^2$ for $j \neq i$, and $\hat{b}_i^2 = 2 - 2 \operatorname{Re} \langle x_i, T_N \cdots T_1 x_i \rangle$.

Embedding. Let $V\xi = 3^{-N/2} \sum_g f(g)\xi \otimes |g\rangle$, let C_i shift the i -th coordinate of $\ell^2(\mathbb{Z}_3^N)$ by -1 , let F_i negate it, and put $X_i = x_i \otimes F_i$. The C_i commute, $X_i C_i X_i = C_i^{-1}$, $X_i C_j X_i = C_j$ for $j \neq i$, and $D^{-1/2} \|C_i V - V c_i\|_{HS} = \hat{a}_i$, $D^{-1/2} \|X_i V - V x_i\|_{HS} = \hat{b}_i$.

Entropy. Let $\rho = VV^*/D$ and let $\mathcal{T}_i$ average conjugation by the powers of C_i . The twirl moves weight at most $\min\{\frac{2}{3}\hat{a}_i^2, \frac{2}{3}\}$ outside the range of VV^* , into a block of rank at most $2D$, so $S(\mathcal{T}_i \rho) \leq \log D + G(\hat{a}_i^2)$; as in Lemma 4.2 the twirls commute, and the increment of each twirl, the Holevo quantity of the conjugates of the current state, can only decrease under the earlier twirls by data processing for the Holevo quantity, so $\bar{\rho} = \mathcal{T}_1 \cdots \mathcal{T}_N \rho$ has $S(\bar{\rho}) \leq \log D + \sum_i G(\hat{a}_i^2)$.

Sectors. Let p be the joint distribution of the eigenvalue labels $Z_i \in \{1, \omega, \omega^2\}$ of the C_i , the same for ρ and $\bar{\rho}$. Then $S(\bar{\rho}) \geq H(p) \geq \sum_i H(Z_i \mid Z_{-i})$. Let Q_a and P_b be the spectral projections of c_i and C_i . The numbers $r_{ab} = D^{-1} \|P_b V Q_a\|_{HS}^2$ form a joint distribution whose marginals are the spectral distribution of c_i in the normalized trace and that of C_i in ρ . Distinct cube roots of unity are at squared distance 3, so $\hat{a}_i^2 = 3 \Pr(a \neq b)$

and $\Delta_i^2 = 3\Pr(a \neq 1)$, whence $s_i := \Pr(Z_i \neq 1) \geq (\Delta_i^2 - \hat{a}_i^2)_+/3$. Next, the normalized vectorizations of V and of $X_i V x_i^*$ are at squared distance $\hat{b}_i^2$, so their real overlap is $1 - \hat{b}_i^2/2$. Measuring the joint spectral projections of the C_j gives p from the first and its flip $p^{(i)}$ in coordinate i from the second, because X_i exchanges ω and ω^2 in coordinate i and fixes the other labels; by the Cauchy–Schwarz inequality outcome by outcome, $\sum_z \sqrt{p(z)p(z^{(i)})} \geq 1 - \hat{b}_i^2/2$. For $0 \leq t \leq 1$ one has $1 - h_2(t) \leq (1 - 2\sqrt{t(1-t)})/\ln 2$: with $y = 2t - 1$ the difference of $1 - \sqrt{1-y^2}$ and $\ln 2(1 - h_2(t))$ vanishes with its derivative at $y = 0$ and has second derivative $(1-y^2)^{-3/2} - (1-y^2)^{-1} \geq 0$. So for fixed other labels with masses u, w on the two nontrivial values of Z_i , the conditional entropy receives at least $u + w - (u + w - 2\sqrt{uw})/\ln 2$, and summing gives $H(Z_i | Z_{-i}) \geq s_i - \hat{b}_i^2/(2\ln 2)$. The numbers in the last claim give $\frac{0.49^2 - 10^{-4}}{3} - \frac{10^{-4}}{2\ln 2} - G(10^{-4}) > \frac{1}{16}$. $\square$

The theorem isolates the row quantities $\hat{a}_i, \hat{b}_i$ that control the dimension estimate used below.

4.3 From an approximate presentation to dimension

Call $\phi : \{a, b, \alpha\} \rightarrow U(D)$ an e -representation if $\|\phi(r_i) - I\|_2 \leq e$ for $i = 1, \dots, 5$, where ϕ is extended to the free group. If w is null in P with area A , then $\|\phi(w) - I\|_2 \leq Ae$, because w is freely a product of A conjugates of relators and the norm is unitarily invariant.

Proposition 4.5. *Let ϕ be an e -representation with $\|\phi(\gamma) - I\|_2 \geq 1/2$, and let $N \geq 1$ satisfy $750\sqrt{N} A_N e \leq 1$, where $A_N = \max\{1, A^*(3N)\}$. Then $\log D \geq N/16$.*

Proof. Put $A_i = \phi(\sigma_{3(i-1)})$ and $B_i = \phi(\sigma_{3(i-1)+1})$ for $1 \leq i \leq N$; all indices are at most $3N - 2$. In H_3 each pair generates a copy of S_3 on $\{(1, 3i-2), (1, 3i-1), (1, 3i)\}$; its involution and braid relations are conjugates of r_1 and r_2 , with defect at most e . Lemma 4.3 gives exact local pairs $x_i = u_i(s)$ and $c_i = u_i(st)$ with $\|x_i - A_i\|_2 \leq e$ and $\|c_i - A_i B_i\|_2 \leq 9e$, and $A_i B_i = \phi(a)^{3(i-1)} \phi(\gamma) \phi(a)^{-3(i-1)}$, so $\Delta_i \geq \frac{1}{2} - 9e$. Pairs with different i have indices differing by at least two, so four and two far-commutator interchanges give $\kappa(c_i, c_j) \leq 4A_N e + 36e \leq 40A_N e$ and $\kappa(x_i, c_j) \leq 2A_N e + 20e \leq 22A_N e$. By Theorem 4.4 and the concavity of G , $\log D \geq N[(\bar{\Delta}^2 - \bar{a})/3 - \bar{b}/(2\ln 2) - G(\bar{a})]$ with $\bar{a} = N^{-1} \sum_i \hat{a}_i^2 \leq \frac{2 \cdot 40^2}{3 \cdot 750^2}$, $\bar{b} = N^{-1} \sum_i \hat{b}_i^2 \leq \frac{4 \cdot 22^2}{3 \cdot 750^2}$ and $\bar{\Delta}^2 \geq (\frac{1}{2} - \frac{9}{750})^2$. Exact rational arithmetic, using $\ln 2 > 69/100$ and $h_2(q) \leq q \log(1/q) + q/\ln 2$, shows that the bracket exceeds $1/16$ by 141043/1397250000. $\square$

If instead $\|\phi(\gamma) - I\|_2 \geq \Delta$ with $0 < \Delta \leq 1$, the same argument with 750 replaced by a constant depending only on Δ gives $\log D \geq c_\Delta N$ with $c_\Delta > 0$. With $A^*(M) \leq CM^p$ the condition reads $N^{p+1/2} e = O(1)$, which gives $\log D \geq c e^{-2/(2p+1)}$: the exponent is $2/(2p+1)$ in $1/e$, or $1/(2p+1)$ in the squared defect $1/e^2$. For $p = \log 130$ this is the exponent $2\beta_H$ of Theorem 1.3.

5 Lower bounds

Proposition 5.1. *Let ϕ assign permutations of a finite set Ω to a, b, α , with $d_n(\phi(r_i), \text{id}) \leq \delta$ for all i and $d_n(\phi(\gamma), \text{id}) \geq 1/2$, and let $N \geq 1$ satisfy $10^8 N A_N \delta \leq 1$, where $A_N = \max\{1, A^*(3N)\}$. Then $\log |\Omega| \geq N/16$.*

Proof. Use the pairs $A_i = \phi(\sigma_{3(i-1)})$, $B_i = \phi(\sigma_{3(i-1)+1})$, $1 \leq i \leq N$, as permutation matrices, with $\|P - I\|_2^2 = 2d_n(P, \text{id})$. Their S_3 relators have $d_n \leq \delta$, hence $\|\cdot\|_2$ -defect at most $e = \sqrt{2\delta}$, and Lemma 4.3 gives exact representations u_i within $10e$ of the normal forms. Commuting two normal forms from different pairs takes at most nine far-commutator interchanges, so by

transport in measure their commutator has $d_n \leq 9A_N\delta$ and $\|\cdot\|_2 \leq 3\sqrt{2A_N\delta}$; rounding adds at most $40e$, so $\eta \leq 43\sqrt{2A_N\delta}$. The element $u_i(st)$ lies within $9e$ of a conjugate of $\phi(\gamma)$, whose $\|\cdot\|_2$ -displacement is at least 1, so $\Delta \geq 1 - 9\sqrt{2\delta} \geq \frac{1}{2}$. Then $t^2 = 2N\eta^2 \leq 7396NA_N\delta < 10^{-4}$, and Lemma 4.2 gives $\log |\Omega| \geq N[(49/100)^2/3 - h_2(10^{-4}) - 10^{-4} \log 5] \geq N/16$. $\square$

With $A^*(M) \leq CM^p$ the condition reads $N^{p+1}\delta = O(1)$, which gives the sofic exponent $1/(p+1)$.

Proof of Theorem 1.1. Let $f : E_H \rightarrow \text{Sym}_n$ be a $(1 - r^{-1})$ -expansive r^{-1} -morphism with $r \geq 10$, and regard permutations as unitary matrices. For permutation matrices, $\|P - P'\|_2^2 = 2d_n(P, P')$, and d_n is bi-invariant. Put $\phi(s) = f(s)$ for $s \in \{a, b, \alpha\}$ and extend ϕ to the free group. Since $s \cdot s^{-1} = 1$ in the chunk, $d_n(f(s)^{-1}, f(s^{-1})) \leq r^{-1}$. For a relator $s_1 \cdots s_l$ with prefixes p_k , the chunk contains $p_k s_{k+1} = p_{k+1}$ and $p_l = 1$, so $l - 1$ merges and at most l inverse replacements give $d_n(\phi(r_i), \text{id}) \leq (2l - 1)/r \leq 23/r$, that is $\|\phi(r_i) - I\|_2 \leq \sqrt{46/r}$. In the same way $d_n(\phi(\gamma), f(\gamma)) \leq 4/r$, while expansiveness gives $d_n(f(\gamma), \text{id}) \geq 1 - 1/r$ because $\gamma \neq 1$; so $d_n(\phi(\gamma), \text{id}) \geq 1 - 5/r \geq 1/2$. Apply Proposition 5.1 with $\delta = 23/r$ and $A^*(3N) \leq 30(3N)^{\log 130}$ (Theorem 3.2): the condition $10^8 N \cdot 30(3N)^{\log 130} \delta \leq 1$ holds for $N = \lfloor (r/(23 \cdot 10^8 \cdot 30 \cdot 3^{\log 130}))^{1/(\log 130+1)} \rfloor$. Therefore, whenever $N \geq 1$ (which forces $r \geq 10$),

$$\log \sigma_{E_H}(r) \geq \frac{N}{16} \geq c_S r^{1/(\log 130+1)} - \frac{1}{16},$$

with $c_S = (23 \cdot 10^8 \cdot 30 \cdot 3^{\log 130})^{-1/(\log 130+1)}/16 > 1/940$, and for smaller r the right side is negative. For unitary approximations Proposition 4.5 gives the corresponding bound: by Theorem 3.2, with $K' = 750 \cdot 30 \cdot 3^{\log 130}$ and $N = \lfloor (K'e)^{-2\beta_H} \rfloor$, every e -representation of P on $\mathbb{C}^D$ with $\|\phi(\gamma) - I\|_2 \geq 1/2$ has $\log D \geq (K'e)^{-2\beta_H}/16 - 1/16$, where $\beta_H = 1/(2 \log 130 + 1)$; this is Theorem 1.3. The upper bound $\log \sigma_{E_H}(r) = O(\sqrt{r} \log r)$ is Proposition 6.2. $\square$

6 Upper bounds

6.1 Permutation models

The upper bounds use permutation models that follow the quotient $\mathbb{Z}^2$ on a torus of clock positions and fail at a single corner. Fix integers $M \geq 2$ and $s \geq 2$, put $\ell = 2M + 2$ and $L = 3\ell$, and let $\Omega_{M,s} = (\mathbb{Z}/M\mathbb{Z})^2 \times [s]^L$. For $2 \leq r \leq L$ let $c_r = (1 \ r \ r-1 \ \cdots \ 2) \in \text{Sym}(L)$ be the reverse cycle and $\tau = (1 \ 2)$; products of permutations are read from left to right, and a permutation π of the coordinates acts on colorings by $(x \cdot \pi)(t) = x(t\pi^{-1})$. For $0 \leq i, j < M$ put $A_{i,j} = c_{\ell-i-j}$ and, for $j < M - 1$, $B_{i,j} = c_{2\ell-j}$. On the top row put $B_{i,M-1} = D_i$, where

$$D_0 = c_{2\ell-M+1}, \quad D_{i+1} = c_{\ell-i-(M-1)}^{-1} \tau D_i c_{\ell-i} \quad (0 \leq i < M - 1). \quad (2)$$

Define permutations of $\Omega_{M,s}$, with clock coordinates modulo M , by

$$(i, j, x) \cdot a = (i + 1, j, x \cdot A_{i,j}), \quad (i, j, x) \cdot b = (i, j + 1, x \cdot B_{i,j}), \quad (i, j, x) \cdot \alpha = (i, j, x \cdot \tau).$$

Proposition 6.1 (Corner models). *In these permutations r_1, r_2, r_3, r_5 hold exactly and r_4 fails only at the clock $(M-1, M-1)$, where it acts as the product $R_M = (1 \ M+4) \prod_{t=5}^{M+3} (t \ t+M)$ of M disjoint transpositions of the colours; so every relator moves at most a fraction M^{-2} of $\Omega_{M,s}$, and γ moves a fraction $1 - s^{-2}$. For $M \geq 5$ the far commutator $C_{2M-1} = [\alpha, a^{2M-1} \alpha a^{1-2M}]$ moves a fraction at least $(1 - s^{-2})(M-4)(M-3)/(2M^2)$. A word of length at most t that represents a nonidentity element of H_3 fixes at most a fraction $s^{-1} + 4t/M$ of $\Omega_{M,s}$ when $M > 2t$.*

Proof. Cycle identities. For $r \geq 4$, $c_r \tau c_r^{-1} = (2\ 3)$ and $c_r (2\ 3) c_r^{-1} = (3\ 4)$, and for $3 \leq r < t$

$$c_r c_t c_{r-1}^{-1} c_t^{-1} = \tau; \quad (3)$$

both sides exchange 1 and 2 and fix every other point. By induction on i , the recursion (2) gives $D_i(t) = t - 1$ for $2 \leq t \leq M + 3 - i$: this holds for the reverse cycle D_0 , and a point $t \leq M + 2 - i$ passes the four factors of D_{i+1} as $t \mapsto t + 1 \mapsto t + 1 \mapsto t \mapsto t - 1$. Since $M + 3 - i \geq 4$, $D_i^{-1}(1) = 2$ and $D_i^{-1}(2) = 3$, so $D_i \tau D_i^{-1} = (2\ 3)$.

Relators. At every clock the colour permutation of aaa^{-1} is $(2\ 3)$, that of $a^2 \alpha a^{-2}$ is $(3\ 4)$, and that of bab^{-1} is $(2\ 3)$, on the top row by the property of D_i . Hence r_1, r_2, r_3, r_5 act trivially everywhere, and γ acts at every clock as a three-cycle of coordinates, which fixes a fraction s^{-2} of the colorings. At the clock (i, j) the colour permutation of r_4 is $A_{i,j} B_{i+1,j} A_{i,j+1}^{-1} B_{i,j}^{-1} \tau$. For $j < M - 1$ it is trivial by (3) with $r = \ell - i - j$ and $t = 2\ell - j$, also for $i = M - 1$ because $B_{i,j}$ does not depend on i off the top row. For $j = M - 1$ and $i < M - 1$ it is trivial by (2). Only the clock $(M - 1, M - 1)$ remains. There the colour permutation is $c_4 D_0 c_{M+3}^{-1} D_{M-1}^{-1} \tau$. With $E_r = c_r^{-1} \tau = (2\ 3 \cdots r)$ the recursion unrolls to $D_{M-1} = E_5 E_6 \cdots E_{M+3} D_0 c_{2M+2} c_{2M+1} \cdots c_{M+4}$, and $D_0 c_r^{-1} D_0^{-1} = E_{r+1}$ for $r \leq 3M + 4$, so the corner permutation is $c_4 H G^{-1} \tau$ with $H = E_{M+4} \cdots E_{2M+3}$ and $G = E_5 \cdots E_{M+3}$. Following the points, H sends $t \mapsto t + M$ for $2 \leq t \leq M + 3$ and $t \mapsto 2M + 5 - t$ for $M + 4 \leq t \leq 2M + 3$, and G sends $t \mapsto t + M - 1$ for $2 \leq t \leq 4$ and $t \mapsto M + 5 - t$ for $5 \leq t \leq M + 3$, both fixing all other points. Hence $HG^{-1} = (2\ 3\ 4\ M+4) \prod_{t=5}^{M+3} (t\ t+M)$, and $c_4 H G^{-1} \tau = R_M$.

Far commutators. Let $k = 2M - 1$, fix a clock (i, j) with $i + j \geq M + 3$ (so $i \geq 4$), and put $F = \ell - j$. The colour permutation of a^k read from this clock is $P = \prod_{t=0}^{k-1} c_{F - ((i+t) \bmod M)}$, and that of σ_k is $P \tau P^{-1}$, which moves $P^{-1}(1)$ and $P^{-1}(2)$. With $d_r = c_r^{-1} = (1\ 2 \cdots r)$,

$$x \cdot (d_r d_{r+1} \cdots d_t) = \begin{cases} x + t - r + 1, & x < r, \\ t - x + 1, & r \leq x \leq t, \\ x, & x > t. \end{cases}$$

The factors of P^{-1} come in three blocks, with cycle lengths $F - i + 2, \dots, F$, then $F - M + 1, \dots, F$, then $F - M + 1, \dots, F - i$. The first block sends $(1, 2)$ to $(i, i + 1)$; since $i \geq F - M + 1$, the second sends $(i, i + 1)$ to $(F - i + 1, F - i)$; the third fixes $F - i + 1$ and sends $F - i$ to 1. So σ_k acts as the transposition $(1\ 2M + 3 - i - j)$, whose second point is at least 5, and C_k acts as a three-cycle. There are $(M - 4)(M - 3)/2$ such clocks.

Short words. Let $\mathcal{S}_{i,j}$ consist of the first $\ell - i - j$, $\ell + i$ and $\ell + j$ points of rays 1, 2, 3, ordered ray by ray; it has L points. When the clock step does not wrap, the generator a of H_3 maps $\mathcal{S}_{i,j}$ onto $\mathcal{S}_{i+1,j}$ and induces $c_{\ell-i-j}$ on the ordered coordinates; likewise b induces $c_{2\ell-j}$ and α induces τ . So on the clocks at distance at least t from the lines where the clock wraps, a fraction at least $1 - 4t/M$ when $M > 2t$, a word of length at most t acts through the element of H_3 it represents. If that element has nonzero translation it fixes no state. If it is nontrivial with zero translation, it moves only points among the first $t + 2$ of each ray, which lie in $\mathcal{S}_{i,j}$, so it acts as a nonidentity permutation of the coordinates and fixes at most a fraction s^{-1} of the colorings. Its fixed-point fraction is therefore at most $s^{-1} + 4t/M$. $\square$

Proposition 6.2 (Upper bound for every fixed chunk). *For every finite chunk $E \subset H_3$, $\log \sigma_E(r) = O_E(\sqrt{r} \log r)$. In particular this holds for E_H .*

Proof. Fix representative words for the elements of E , with the empty word representing 1, and let t bound the lengths of their pairwise differences. The finitely many chunk relations have some maximum area B_E in Johnson's presentation. Take $s = M$ and the diagonal action

on two independent copies of $\Omega_{M,M}$. For $M > 4t$, the short-word estimate in Proposition 6.1 bounds the fixed-point fraction of every nonidentity difference by $((1+8t)/M)^2$ in the product: the fractions square. Each relator has defect at most $2/M^2$ in the product, so transport in measure bounds every chunk-relation defect by $2B_E/M^2$. Choosing M at least a sufficiently large E -dependent multiple of $\sqrt{r}$ gives an r^{-1} -morphism that is $(1-r^{-1})$ -expansive. Its degree is $|\Omega_{M,M}|^2 = M^{4+2(6M+6)}$, whose logarithm is $O(M \log M)$, proving the claim. $\square$

6.2 Unitary models

In a permutation model a relator that fails costs the fraction of points it moves. In a unitary model it costs a sum of squared angles, so a single defect can be spread thin. Doing this for the corner models gives unitary models whose relator defect has squared norm of order M^{-3} , not M^{-2} , at the same size. Throughout, products are read from left to right, the left factor acting first, as in Section 6, and $e_1 = (1, 0)$, $e_2 = (0, 1)$.

One-particle models. No generator moves the last $2M+2$ of the $6M+6$ colour coordinates. Use the reduced colour coordinates $1, \dots, L$ with $L = 4M+4$, and read each link permutation A_c , B_c and τ of Section 6 as a permutation matrix on $\mathbb{C}^L$. A one-particle model replaces some of these link matrices by unitaries. For a word w and a clock c the holonomy $W_c(w) \in U(L)$ is the product of the link matrices along the path of w from c , with inverse matrices for inverse letters. For a two-element set $S = \{x, y\}$ let Q_S be the projector onto $(e_x - e_y)/\sqrt{2}$; then $\exp(i\pi Q_S)$ is the transposition $(x\ y)$, and $\exp(i\vartheta Q_S)$ is a phase rotation in the plane of S .

Fermionic quantization. For $W \in U(L)$ let $\Lambda(W)$ act on the exterior algebra $\Lambda(\mathbb{C}^L)$, of dimension 2^L , by $v_1 \wedge \dots \wedge v_m \mapsto Wv_1 \wedge \dots \wedge Wv_m$. It is multiplicative on all of $U(L)$, and $\text{Tr } \Lambda(W) = \det(I + W)$, since in degree m its eigenvalues are the products of m eigenvalues of W . For $k \geq 1$ put $\Lambda_k = \Lambda^{\otimes k}$, so that $2^{-kL} \text{Tr } \Lambda_k(W) = \det((I + W)/2)^k$. The quantized model acts on $\mathcal{H} = \mathbb{C}^{M^2} \otimes \Lambda(\mathbb{C}^L)^{\otimes k}$, with normalized trace τ . A letter moves the clock c to $c + \pi(s)$ and applies Λ_k of its link matrix at c , and $\rho(s)$ is the adjoint of this operator, so ρ is a representation of the free group on a, b, α . A word whose translation is nonzero modulo M has $\tau(\rho(w)) = 0$, and a word that closes has

$$\|\rho(w) - I\|_2^2 = \mathbb{E}_c \left[2 - 2 \text{Re} \det((I + W_c(w))/2)^k \right],$$

with $\mathbb{E}_c$ the average over clocks. For a three-cycle σ of coordinates $\det((I + \sigma)/2) = \frac{1}{4}$.

Lemma 6.3 (Rotating the links). *In a corner model, replace every link A_c by $U_c A_c$ and every link B_c by $V_c B_c$, where U_c commutes with $(2\ 3)$ and $(3\ 4)$, and V_c commutes with $(2\ 3)$. Then at every clock the holonomies of α , $a\alpha a^{-1}$, $a^2\alpha a^{-2}$ and $b\alpha b^{-1}$ are $(1\ 2)$, $(2\ 3)$, $(3\ 4)$ and $(2\ 3)$, so r_1, r_2, r_3, r_5 have identity holonomy everywhere and γ acts as a three-cycle.*

Proof. By the proof of Proposition 6.1, $A_c \tau A_c^{-1} = (2\ 3)$, $A_c (2\ 3) A_c^{-1} = (3\ 4)$ and $B_c \tau B_c^{-1} = (2\ 3)$ at every clock. So the holonomy of $a\alpha a^{-1}$ at c is $U_c (2\ 3) U_c^{-1} = (2\ 3)$, that of $a^2\alpha a^{-2}$ is $U_c A_c U_{c+e_1} (2\ 3) U_{c+e_1}^{-1} A_c^{-1} U_c^{-1} = U_c (3\ 4) U_c^{-1} = (3\ 4)$, and that of $b\alpha b^{-1}$ is $V_c (2\ 3) V_c^{-1} = (2\ 3)$. The relators r_1, r_2, r_3, r_5 are products of these conjugates, as in the proof of Proposition 6.1. $\square$

A plaquette q with base v_0 has the vertices v_0 , $v_1 = v_0 + e_1$, $v_2 = v_0 + e_2$, $v_0 + e_1 + e_2$; its bottom and top edges are the a -links at v_0 and v_2 , and its left and right edges the b -links at v_0 and v_1 . Its r_4 holonomy is $W_q = A_{v_0} B_{v_1} A_{v_2}^{-1} B_{v_0}^{-1} \tau$. A *region* consists of a set R of plaquettes, a source $q_0 \in R$, a sign $s_R = \pm 1$, and a two-element label $\text{lab}(v) \subset \{1, \dots, L\}$ at every vertex of every plaquette of R . An edge is *interior* if it lies on two plaquettes of R .

Lemma 6.4 (Spreading a defect). *Take a permutation model satisfying the local identities $A_c \tau A_c^{-1} = (2\ 3)$, $A_c(2\ 3)A_c^{-1} = (3\ 4)$ and $B_c \tau B_c^{-1} = (2\ 3)$ of Lemma 6.3 at every clock. Suppose W_q is the identity except at finitely many sources, where it is the product of the transpositions given by the base labels of the regions with that source. Let finitely many regions satisfy:*

- (R1) *every edge of a non-source plaquette, from v to w , carries $\text{lab}(v)$ onto $\text{lab}(w)$ under its link permutation;*
- (R2) *every base label avoids $\{1, 2\}$, and the label at the initial vertex of every interior edge avoids $\{2, 3, 4\}$;*
- (R3) *each region is connected through its interior edges, and the only interior edge of its source is the bottom edge;*
- (R4) *labels of different regions are disjoint wherever both are defined, and disjoint from any other modification of the link.*

Let $F_{R,q}$ be real numbers with $\sum_{q \in R} F_{R,q} = s_R \pi$ for each region. Then there are potentials φ_R on the interior edges of R such that the model with $U_c = \prod_R \exp(i\varphi_R(a\text{-edge at } c) Q_{\text{lab}_R(c)})$ and $V_c = \prod_R \exp(i\varphi_R(b\text{-edge at } c) Q_{\text{lab}_R(c)})$ still has r_1, r_2, r_3, r_5 exact, and its r_4 holonomy at every plaquette q is $\prod_{R \ni q} \exp(iF_{R,q} Q_{\text{lab}_R(v_0)})$, with v_0 the base of q .

Proof. Put $\varphi_R = 0$ off the interior edges and $(\text{curl } \varphi)(q) = \varphi(\text{bottom}) + \varphi(\text{right}) - \varphi(\text{top}) - \varphi(\text{left})$. An interior edge enters the curls of its two plaquettes with opposite signs, so the curls over R sum to zero. Choose a spanning tree of the plaquettes of R joined through interior edges, rooted at the source, and fix the potential of each tree edge, from the leaves inward, so that $\text{curl } \varphi_R = F_R - s_R \pi \delta_{q_0}$ at the plaquette below it; since both sides sum to zero, the equation then also holds at the root. By (R2) and (R4) all rotations on a link commute with the transpositions of Lemma 6.3, so r_1, r_2, r_3, r_5 stay exact.

For the r_4 holonomy, the unmodified identity $A_{v_0} B_{v_1} A_{v_2}^{-1} B_{v_0}^{-1} = W_q \tau$ gives

$$W'_q = U_{v_0} (A_{v_0} V_{v_1} A_{v_0}^{-1}) W_q \tau (B_{v_0} U_{v_2} B_{v_0}^{-1})^{-1} V_{v_0}^{-1} \tau.$$

Conjugation by a link permutation moves the plane of a rotation, and at a non-source plaquette (R1) moves the planes $\text{lab}(v_1)$ and $\text{lab}(v_2)$ to $\text{lab}(v_0)$. At a source only the bottom edge carries a potential. So all rotations in W'_q lie in the planes $\text{lab}_R(v_0)$; they avoid $\{1, 2\}$ by (R2), hence commute with τ , and planes of different regions are disjoint by (R4). For region R the angles add up to $(\text{curl } \varphi_R)(q)$, and at its source W_q contributes the rotation by $s_R \pi$ in the plane of the source label. The total is $F_{R,q}$. $\square$

Proposition 6.5 (Unitary models). *For every $M \geq 48$ the model above with $k = 2$ and suitable rotations of the links gives unitaries $\rho(a), \rho(b), \rho(\alpha)$ on a space of dimension $M^2 2^{8M+8}$ in which r_1, r_2, r_3, r_5 act as the identity, $\|\rho(r_4) - I\|_2^2 \leq 14.26 M^{-3}$ and $\|\rho(\gamma) - I\|_2^2 = 15/8$.*

Proof. Two exact moves. Let $c^* = (M-1, M-1)$, where the r_4 holonomy is R_M (Proposition 6.1). Replace A_{c^*} by $\sigma_1 A_{c^*}$ and $B_{(0, M-2)}$ by $\sigma_2 B_{(0, M-2)}$, with $\sigma_1 = (1\ M+4)$ and $\sigma_2 = (5\ M+5)$. These satisfy the hypotheses of Lemma 6.3, so r_1, r_2, r_3, r_5 stay exact. The modified links lie on three plaquettes. At c^* the holonomy becomes $\sigma_1 R_M = \prod_{t=5}^{M+3} (t\ t+M)$. At $(0, M-2)$, where it was the identity, it becomes $\tau \sigma_2 \tau = \sigma_2$. At $(M-1, M-2)$ it becomes $(A \sigma_2 A^{-1}) \tau (B \sigma_1 B^{-1}) \tau$ with $A = A_{(M-1, M-2)} = c_5$ and $B = B_{(M-1, M-2)} = c_{3M+6}$. Here $A \sigma_2 A^{-1} = (1\ M+5)$ and $B \sigma_1 B^{-1} = (2\ M+5)$, so $\tau (B \sigma_1 B^{-1}) \tau = (1\ M+5)$ and the holonomy stays the identity.

The regions. For $5 \leq t \leq M+3$ let R_t have source c^* and sign $(-1)^{t-5}$, and consist of c^* , the plaquettes $(M-1, j)$ with $1 \leq j \leq M-2$, and the plaquettes (i, j) with $0 \leq i \leq M-3$,

$0 \leq j \leq M-2$ and $i+j \leq t+M-8$; label the vertex (i, j) by $\{t+M-1-i-j, t+2M-1-j\}$ for $0 \leq i \leq M-2$ and $(M-1, j)$ by $\{t+M-1-j, t+2M-1-j\}$. Let T have source $(0, M-2)$ and sign $(-1)^{M-1}$, and consist of $(0, M-2)$ and the plaquettes (i, j) with $i, j \geq 0$ and $i+j \leq M-3$, with labels $\{M+3-i-j, 2M+3-j\}$. The source labels are the defect pairs $\{t, t+M\}$ at c^* and $\{5, M+5\}$ at $(0, M-2)$.

These regions satisfy (R1)–(R4). Off the top row $B_{i,j} = c_{4M+4-j}$ lowers every label coordinate by one, which is the change of the labels from j to $j+1$. In the triangles $A_{i,j} = c_{2M+2-i-j}$ lowers the first label coordinate, which is at most $2M+2-i-j$ since $t \leq M+3$, and fixes the second, which exceeds $2M+2-i-j$ since $t \geq 5$; this is the change from i to $i+1$, and the same holds in T . On the column $i = M-1$, $A_{M-1,j} = c_{M+3-j}$ fixes both label coordinates, which are at least $t+M-1-j \geq M+4-j$, and the labels at $(M-1, j)$ and $(0, j)$ agree; this is the only crossing of a seam, since the rows stay below $M-1$ and c^* is attached only through its bottom edge. Every base label and every label at the initial vertex of an interior edge has both coordinates at least 5. Two labels of bulk regions at one vertex differ in t by at most $M-2$, so they are disjoint; a bulk label meets a label of T only if $t \in \{4, 4-M-i, M+4+i\}$, which is impossible; and σ_1, σ_2 are disjoint from the labels $\{t, t+M\}$ and $\{t+1, t+M+1\}$ on the links they modify. Each region is connected through interior edges: the triangle is a staircase, the column is glued to it through the b -edges at $(0, j)$, and each source through its bottom edge. Finally $R_t \subset R_{t+1}$, and every region has at least $m_M = (M-1)(M-2)/2 + 1$ plaquettes.

The defect. Take $F_{R,q} = s_R \pi / |R|$ on each region and $k = 2$. By Lemma 6.4 the r_4 holonomy at q is a product of phase rotations by angles $F_{R,q}$ in disjoint planes, so $\det((I + W'_q)/2) = \prod_R (1 + e^{iF_{R,q}})/2 = e^{i\Theta_q} C_q$ with $\Theta_q = \frac{1}{2} \sum_R F_{R,q}$ and $C_q = \prod_R \cos(F_{R,q}/2) \in [0, 1]$. Using $1 - C^k \leq k(1 - C)$, $1 - \prod_R c_R \leq \sum_R (1 - c_R)$ and $1 - \cos x \leq x^2/2$,

$$2 - 2 \operatorname{Re}(e^{ik\Theta_q} C_q^k) = 2(1 - C_q^k) + 2C_q^k(1 - \cos k\Theta_q) \leq \frac{k}{4} \sum_R F_{R,q}^2 + k^2 \Theta_q^2.$$

Summed over the plaquettes, $\sum_q \sum_R F_{R,q}^2 = \sum_R \pi^2 / |R| \leq M\pi^2 / m_M$. The bulk regions containing q are those with t at least some t_q ; their signs alternate and their magnitudes $\pi / |R_t|$ decrease, so they sum to at most π / m_M in absolute value, and T adds at most π / m_M . Hence $|\Theta_q| \leq \pi / m_M$, and averaging over the M^2 clocks,

$$M^3 e_4^2 := M^3 \|\rho(r_4) - I\|_2^2 \leq \frac{\pi^2 M^2}{2m_M} + \frac{4\pi^2 M^3}{m_M^2} \leq \frac{\pi^2 M^2}{(M-1)(M-2)} + \frac{16\pi^2 M^3}{(M-1)^2(M-2)^2},$$

which decreases in M and is below 14.26 at $M = 48$. So $e_4^2 \leq 14.26 M^{-3}$ for $M \geq 48$.

The three-cycle. By Lemma 6.3, γ acts at every clock as a three-cycle σ of coordinates, so $\tau(\rho(\gamma)) = \det((I + \sigma)/2)^2 = 4^{-2}$ and $\|\rho(\gamma) - I\|_2^2 = 2 - 2 \cdot 4^{-2} = 15/8$. $\square$

Corollary 6.6 (Unitary upper bound). *Let $0 < e \leq e_0 = 14.26^{1/2} \cdot 48^{-3/2}$ and $M = \lceil (14.26/e^2)^{1/3} \rceil$. Johnson's presentation has an e -representation on $\mathbb{C}^D$ with $\|\phi(\gamma) - I\|_2 \geq 1/2$ and $\log D \leq 8M + 8 + 2 \log M$. So such representations exist with $\log D = O(e^{-2/3})$.*

Proof. The choice of M gives $M \geq 48$ and $14.26 M^{-3} \leq e^2$; apply Proposition 6.5. $\square$

Our permutation construction gives the weaker bound in this norm. Read as permutation matrices, the corner models with $s = 2$ have relator defects at most $\sqrt{2}/M$ in $\|\cdot\|_2$, since a permutation moving a fraction f of the points has $\|P - I\|_2^2 = 2f$, and $\|\phi(\gamma) - I\|_2^2 = 3/2$, at $\log D = 6M + 6 + 2 \log M$; this gives only $\log D = O(e^{-1})$. Spreading the defect lowers the exponent of $1/e$ in our upper bounds from 1 to $\frac{2}{3}$; whether optimal permutation models are weaker than optimal unitary ones is open (Section 8). The lower bound of Theorem 1.3 has exponent $2\beta_H \approx 0.133$.

7 The correction

Corollary 7.1. *Theorem 3.15 of [2] is false for $1 \rightarrow \text{FSym}(\mathbb{N}) \rightarrow H_3 \rightarrow \mathbb{Z}^2 \rightarrow 1$ and the chunk E_H . Consequently the conclusion of [2, Example 3.16], that every elementary amenable group has polynomial sofic profile, fails for H_3 .*

Proof. Theorem 3.15 of [2] asserts a symmetric chunk E' of the kernel and a finite symmetric $S \subset \mathbb{Z}^2$ with $\sigma_{E_H}(r) \leq \sigma_{E'}(r) \alpha_{\mathbb{Z}^2, S}(r)$ for all $r > 1$, where $\alpha_{Q, S}(r)$ is the least size of a finite $F \subset Q$ with $|SF \setminus F| < |F|/r$. The kernel $\text{FSym}(\mathbb{N})$ is locally finite, so E' lies in a finite subgroup and $\sigma_{E'}$ is bounded [2, Fact 3.5]; boxes show $\alpha_{\mathbb{Z}^2, S}(r) = O(r^2)$. The resulting bound $\sigma_{E_H}(r) = O(r^2)$ contradicts Theorem 1.1. Since H_3 is elementary amenable, Example 3.16 fails. The contradiction does not need Theorem 3.2: with the certified bound of Theorem 3.1 alone, the proof of Theorem 1.1 gives $\sigma_{E_H}(r) \geq 2^{r^{1/12}/270-1/16}$. $\square$

Where the argument breaks. Theorem 3.15 is stated in [2] as what the explicit construction of Elek and Szabó [3] yields “without any change”. That construction, given a finite $F \subset G$ and $\varepsilon > 0$, chooses a Følner set $A \subset Q$ with $|Ag \setminus A| \leq \varepsilon|A|$ for $g \in F$, a section $\sigma : Q \rightarrow G$, and then an approximation of the kernel on the finite set

$$H_\varepsilon = N \cap (\sigma(A) F \sigma(A)^{-1}),$$

acting on $B \times A$ with B the kernel’s approximation space [3, proof of Theorem 1(3)]. The kernel chunk is therefore H_ε , and it depends on $\varepsilon = 1/r$ through A . For H_3 with the section $(m, n) \mapsto a^m b^n$ and $A = [0, M]^2$, it contains $a^m b^n \alpha b^{-n} a^{-m}$, the transposition of $(1, m+n+1)$ and $(1, m+n+2)$, for all $0 \leq m, n < M$: at least $2M-1$ distinct transpositions, which generate the symmetric group of $2M$ points, with M of order r . No chunk independent of r contains them. What the construction does give is the bound of Proposition 7.2 below, in which the kernel chunk grows with r . The upper bound by the isoperimetric profile [2, Proposition 3.14] is not affected, and H_3 , having exponential growth, has isoperimetric profile at least exponential. The proof of [2, Proposition 3.17] for Baumslag–Solitar groups applies Theorem 3.15 in the same way, so it is incomplete as well; we do not know whether its conclusion holds.

A scale-dependent extension bound. The construction of Elek and Szabó, with the accuracy tracked explicitly, gives the following correct form of the extension bound. Let $1 \rightarrow N \rightarrow G \xrightarrow{\pi} Q \rightarrow 1$ be an extension, $s : Q \rightarrow G$ a section with $s(e) = 1$, and put $c(a, g) = s(a) g s(a\pi(g))^{-1} \in N$ for $a \in Q$ and $g \in G$.

Proposition 7.2. *Let $E \subset G$ be a symmetric chunk, $r > 1$, and $A \subset Q$ a finite nonempty set with $|A\pi(g) \setminus A| \leq |A|/(4r)$ for every $g \in E$. Then $K_{A, E} = \{c(a, g) : g \in E, a \in A, a\pi(g) \in A\}$ is a symmetric chunk of N , and*

$$\sigma_E(r) \leq |A| \sigma_{K_{A, E}}(2r).$$

Proof. The set $K_{A, E}$ contains $c(a, 1) = 1$, and $c(a, g)^{-1} = c(a\pi(g), g^{-1})$, so it is symmetric. The cocycle identity $c(a, g) c(a\pi(g), h) = c(a, gh)$ holds in G . Let $f_K : K_{A, E} \rightarrow \text{Sym}_m$ be a $(1 - \frac{1}{2r})$ -expansive $\frac{1}{2r}$ -morphism with $m = \sigma_{K_{A, E}}(2r)$, and on $A \times [m]$ let $f(g)$ send (a, x) to $(a\pi(g), x \cdot f_K(c(a, g)))$ whenever $a\pi(g) \in A$, completed arbitrarily to a permutation; $f(1)$ is the identity. Call a fiber a good for a relation $gh = k$ in E if $a, a\pi(g)$ and $a\pi(k)$ lie in A . Then $c(a, g) c(a\pi(g), h) = c(a, k)$ is a relation of the chunk $K_{A, E}$, so on a good fiber $f(g)f(h)$ and $f(k)$ differ on at most a fraction $\frac{1}{2r}$ of the points, and the bad fibers form a fraction at most $\frac{2}{4r}$. So f is an r^{-1} -morphism. For distinct $g, h \in E$: if $\pi(g) \neq \pi(h)$, then on every

fiber a with $a\pi(g), a\pi(h) \in A$ the two images have different quotient coordinates, a fraction at least $1 - \frac{1}{2r}$ of the points; if $\pi(g) = \pi(h)$, then $c(a, g) \neq c(a, h)$ are distinct elements of $K_{A,E}$, so on the fibers with $a\pi(g) \in A$ the images differ on a fraction at least $1 - \frac{1}{2r}$ of each fiber, and in total on at least $(1 - \frac{1}{4r})(1 - \frac{1}{2r}) \geq 1 - \frac{1}{r}$. So f is $(1 - r^{-1})$ -expansive, on $|A| m$ points. $\square$

When the cocycle takes finitely many values on $Q \times E$, for instance for direct products, $K_{A,E}$ lies in a fixed finite set and a bound with a fixed kernel chunk follows, as Theorem 3.15 asserts. For H_3 with boxes $A = [0, M)^2$ and M of order r , K_{A,E_H} lies in the finite group of permutations of the first $2M + O(1)$ points of ray 1, whose regular representation gives $\sigma_{K_{A,E_H}} \leq (2M + O(1))!$; so the proposition gives $\log \sigma_{E_H}(r) = O(r \log r)$, consistent with, and weaker than, Proposition 6.2.

All numbering refers to the published version [2]. Theorem 3.15 and Example 3.16 read the same there as in the arXiv version, where Proposition 3.17 appears as Example 3.17.

8 Open problems

- (1) *The exponents.* The sofic exponent of E_H in r lies between $1/(\log 130 + 1) \approx 1/8.02$ and $1/2$ (up to the logarithm). For unitary approximations with relator defect e in the normalized Hilbert–Schmidt norm, $\log D$ lies between order $e^{-2\beta_H}$, with $2\beta_H \approx 0.133$ (Theorem 1.3), and $O(e^{-2/3})$ (Corollary 6.6). The exponent is essentially intrinsic to the particular substitution certificates used here: reweighting these fillings cannot go below $\log \rho \approx 7.018$, with $\rho \approx 129.64$ the spectral radius of their substitution matrix, and $\log 130$ is the certified choice. Since the area is at least quadratic (Proposition 3.4), the present uniform-area transport and sector argument cannot give more than $1/3$ in r for permutations and $2/5$ in $1/e$ for unitaries. A quadratic Dehn function for H_3 would reach these values: every word in the argument has linear length. Theorem 3.3 gives the polynomial upper bound $O(n^{\log 130 + 4})$, while for Thompson’s group F a quadratic Dehn function is a theorem [13]. Shared fillings, improved aggregate defect estimates, or a stronger dimension argument are possible routes beyond these limits.
- (2) *Permutation versus unitary approximation.* Relator errors add up in measure for permutations and in norm for unitaries, so the two profiles may differ. The best models known already do: in the normalized Hilbert–Schmidt norm, permutation models of H_3 give $\log D = O(e^{-1})$ and unitary models $O(e^{-2/3})$ (Section 6.2). Can the unitary profile of a chunk of an amenable group be polynomially smaller, in exponent, than its sofic profile?
- (3) *Other groups.* Every Houghton group H_m with $m \geq 3$ contains H_3 ; the same method applies to groups with a displaced S_3 structure and polynomially bounded far-commutator area. The proof of [2, Proposition 3.17] for Baumslag–Solitar groups uses the same extension theorem; whether its conclusion holds is open.
- (4) *Extensions.* Proposition 7.2 bounds the profile of an extension by the Følner size of the quotient and the profile of a kernel chunk that grows with r through the cocycle of a section; the kernel chunk is fixed when the cocycle takes finitely many values. Which extensions of groups with polynomial profile by amenable quotients have polynomial profile?

A Fillings for the far-commutator bound

Areas are taken in P ; free conjugation and inversion do not change them, and freely $[x, yz] = [x, y] \cdot y[x, z]y^{-1}$. Write $\tau_j = b^j \alpha b^{-j}$, $t = b^{-1}a$, $\rho = a^{-1}b^{-1}ab$, $t_i = a^i t a^{-i}$, $X_n = b^{-n}a^n$, and

$$A(d) = \text{Area}[\sigma_0, \sigma_d], \quad E(i) = \text{Area}[\sigma_i, t], \quad U(j) = \text{Area}(\tau_j \sigma_j^{-1}), \quad B(k) = \text{Area}[\sigma_{-k}, b].$$

Since $b^{-n}(\tau_n \sigma_n^{-1})b^n = [\alpha, X_n]$ freely, $U(n) = \text{Area}[\alpha, X_n]$. Since $[x^{-1}, y] = x^{-1}[y, x]x$, the signs in the definition of A^* do not matter and $A^*(M) = \max_{2 \leq d \leq M} A(d)$. Every relation below with a numerical area is certified in the ancillary files (Section 3).

Base relations. $[\alpha, t]$ is a rotation of r_5^{-1} ; $aba^{-1} = \alpha b$ and $bab^{-1} = \alpha^{-1}a$ are rotations of $r_4^{\pm 1}$; $[\sigma_i, \sigma_{i+2}]$ is a conjugate of r_3 ; all have area 1. The relation $B_0 := \text{Area}[\alpha, \sigma_1 \sigma_0 b] \leq 5$ follows by replacing $b\alpha^{-1}b^{-1}$ with σ_1^{-1} (one r_5), which leaves $\sigma_0 \sigma_1 \sigma_0 \sigma_1^{-1} \sigma_0^{-1} \sigma_1^{-1}$, then three r_1 cells and one r_2 cell.

Lemma A.1 (Transfer and halving). *For $i \geq 1$, $j \geq 1$, $k \geq 2$:*

- (i) $E(i) \leq 2i + 1 + \sum_{d=2}^{i+1} A(d)$, and $E(0) = 1$;
- (ii) $U(j) \leq \sum_{i=0}^{j-1} E(i)$;
- (iii) $B(k) \leq 2k + 5 + \sum_{d=2}^{k-1} A(d)$;
- (iv) $A(k+j) \leq A(k) + 2jB(k) + 2U(j)$.

Proof. (i) By $bab^{-1} = \alpha^{-1}a$, $t_k = t_{k+1}\sigma_{k-1}$ with one cell, so $t = t_i \sigma_{i-2} \cdots \sigma_0 \sigma_{-1}$ with i cells, and i more for t^{-1} . Then $[\sigma_i, t_i S] = [\sigma_i, t_i] \cdot t_i[\sigma_i, S]t_i^{-1}$ with $S = \sigma_{i-2} \cdots \sigma_{-1}$; the first factor is $a^i[\alpha, t]a^{-i}$, and the second is a product of conjugates of $[\sigma_i, \sigma_l]$ for $l = i-2, \dots, -1$, of widths $2, \dots, i+1$. (ii) Freely $\tau_{i+1} = b\tau_i b^{-1}$; replace τ_i by σ_i at cost $U(i)$, and use that $b\sigma_i b^{-1} \sigma_{i+1}^{-1} = b[\sigma_i, t]b^{-1}$ freely. (iii) Freely $[\sigma_{-k}, b] = a^{-k}[\alpha, a^k b a^{-k}]a^k$, and $a^k b a^{-k} = \sigma_{k-1} \cdots \sigma_1 \sigma_0 b$ with k cells (twice, for the element and its inverse); then move α past $\sigma_{k-1}, \dots, \sigma_2$ and finish with B_0 . (iv) Conjugating by a^{-k} turns $[\sigma_0, \sigma_{k+j}]$ into $[\sigma_{-k}, \sigma_j]$. Replace $\sigma_j^{\pm 1}$ by $\tau_j^{\pm 1}$ ($2U(j)$), move $\sigma_{-k}^{\pm 1}$ across the $2j$ letters $b^{\pm 1}$ ($2jB(k)$), and what remains is $b^j[\sigma_{-k}, \alpha]b^{-j}$, a conjugate of $[\sigma_0, \sigma_k]$. $\square$

The lemma gives, in this order: $A(2) = 1$, $E(0) = U(1) = 1$, $E(1) \leq 4$, $U(2) \leq 5$, $B(2) \leq 9$, $A(3) \leq 21$ ($k=2, j=1$), $A(4) \leq 47$ ($k=j=2$), $E(2) \leq 27$, $U(3) \leq 32$, $B(3) \leq 12$, $A(5) \leq 79$ ($k=3, j=2$), $A(6) \leq 157$ ($k=j=3$) and $E(3) \leq 76$. Part (iv) with $k=2$ gives $A(m) \leq 2U(m-2) + 18m - 35$ for $m \geq 3$.

The cheap letters. Freely, $\psi(t) = b^{-2}a^2 = tpt$, $t^b = b^{-2}ab = t\rho$, and $\psi(\rho) = \rho^a \rho^{ba} \rho \rho^b = t^{-1} \rho^b \rho^{b^2} t \cdot \rho \cdot \rho^b$ (from $[x^2, y^2] = (xcx^{-1})(xycy^{-1}x^{-1})c(ycy^{-1})$ with $c = [x, y]$, $x = a^{-1}$, $y = b^{-1}$, and $a = bt$). The one relation needed is

$$\rho^b = R' := tpt^{-1} \cdot \rho \cdot tpt^{-1}, \quad \text{area} \leq 14.$$

Replace each ρ by α^{ba} (four r_4 cells); freely the relation becomes $\alpha^{bab} = \alpha^{b^2} \alpha^{ba} \alpha^{b^2}$, and after conjugating by $(bab)^{-1}$ it reads $\alpha = \alpha^g \alpha^{g'} \alpha^g$ with $g = ba^{-1}b^{-1}$ and $g' = bab^{-1}a^{-1}b^{-1}$. Six cells of $bab^{-1} = \alpha^{-1}a$ turn this into $\alpha = \alpha^{-1} \sigma_1 \alpha \cdot bab^{-1} \cdot \alpha^{-1} \sigma_1 \alpha$; one r_5 cell, two r_1 cells and one r_2 cell finish. Also $[\alpha, t]$ has area 1 and $[\alpha, \rho]$ area at most 5: $\rho \alpha^{-1} \rho^{-1}$ becomes $a^{-1} b^{-1} \sigma_2^{-1} ba$ after one r_5 cell, and $b^{-1} \sigma_2^{-1} b \rightarrow \sigma_1^{-1}$ costs $E(1) \leq 4$. Consequently, for a word

W in $t^{\pm 1}, \rho^{\pm 1}$, $W^b = W(t\rho, R')$ at area $14 \#_{\rho}(W)$, and $\psi(W) = W(t\rho t, R)$ at area $84 \#_{\rho}(W)$, where

$$R = t^{-1} \cdot R' \cdot R'(t\rho, R') \cdot t \cdot \rho \cdot R'$$

uses the relation six times per letter ρ ; $|R'| = 7$ and $|R| = 46$.

Lemma A.2 (The doubling lifts). $\psi(\alpha) = \Pi := \sigma_1\sigma_2\sigma_0\sigma_1$ with 6 cells, and $\text{Area } \psi(r_i) \leq 17, 359, 864, 0, 182$ for $i = 1, \dots, 5$. Hence $\text{Area } \psi(w) \leq 864 \text{Area}(w)$ for every null word w .

Proof. The identity for $[x^2, y^2]$ with $x = a, y = b$ and four r_4 cells ($[a, b] = \alpha$) give $\psi(\alpha) = \sigma_1 \cdot a(bab^{-1})a^{-1} \cdot \alpha \cdot bab^{-1}$, and two r_5 cells give Π . $\psi(r_4)$ is freely trivial. $\psi(r_1) = \psi(\alpha)^2$: after two conversions, Π^2 costs five cells (σ_1^2 , one r_3 swap, then $\sigma_2^2, \sigma_0^2, \sigma_1^2$). $\psi(r_5) = a^2\psi(\alpha)a^{-2} \cdot b^2\psi(\alpha)^{-1}b^{-2}$: after two conversions, $a^2\Pi a^{-2} = \sigma_3\sigma_4\sigma_2\sigma_3$ freely, and $b^2\sigma_i^{-1}b^{-2} \rightarrow \sigma_{i+2}^{-1}$ costs $E(i) + E(i+1)$; over the letters of Π this is $E(0) + 3E(1) + 3E(2) + E(3) \leq 170$. $\psi(r_3) = [\psi(\alpha), a^4\psi(\alpha)a^{-4}]$: after four conversions it is $[\Pi, \sigma_5\sigma_6\sigma_4\sigma_5]$, which splits freely into sixteen conjugates of far commutators of widths 2, 3, 3, 3, 3, 4, 4, 4, 4, 4, 5, 5, 5, 5, 6, costing 840. $\psi(r_2) = (\psi(\alpha) a^2\psi(\alpha)a^{-2})^3$: after six conversions it is the positive word $(\Pi \sigma_3\sigma_4\sigma_2\sigma_3)^3$ of length 24 in $\sigma_0, \dots, \sigma_4$, which represents the identity of $\text{Sym}(6)$. Reduce it by the coset procedure: keep the word in the form $h \cdot c_k$ with $c_k = \sigma_4\sigma_3 \cdots \sigma_k$ and h a word in $\sigma_0, \dots, \sigma_3$; absorbing a letter σ_i costs far commutations for $i \leq k-2$, nothing for $i = k-1$, one cancellation for $i = k$, and far commutations plus one braid move for $i \geq k+1$, and then recurse on h . A braid move $\sigma_i\sigma_{i-1}\sigma_i = \sigma_{i-1}\sigma_i\sigma_{i-1}$ costs four cells (three r_1 , one r_2). Charging each far commutation its actual width, the procedure uses 47 moves at total cost 323. The last claim follows by applying ψ to a product of conjugates of relators. $\square$

Proof of Theorem 3.1. Define words in $t^{\pm 1}, \rho^{\pm 1}$ by $W_1 = t$, $W_{2n} = W_n(t\rho t, R)$ and $W_{2n+1} = W_{2n}(t\rho, R')t$, and let $\text{Rw}(n) = \text{Area}(X_n W_n^{-1})$. Then $\text{Rw}(1) = 0$. Since $X_{2n} = \psi(X_n)$ literally, Lemma A.2 and the cheap-letter relations give $\text{Rw}(2n) \leq 864 \text{Rw}(n) + 84|W_n|$ and $|W_{2n}| \leq 46|W_n|$. Since $X_{2n+1} = X_{2n}^b t$ freely, $\text{Rw}(2n+1) \leq \text{Rw}(2n) + 14 \cdot 46|W_n|$ and $|W_{2n+1}| \leq 7|W_{2n}| + 1$. Per binary digit, $\text{Rw} \leq 864 \text{Rw}' + 728|W'|$ and $|W| \leq 323|W'|$, so with $s = \lfloor \log n \rfloor$,

$$|W_n| \leq 323^s, \quad \text{Rw}(n) \leq 728 \sum_{i < s} 864^{s-1-i} 323^i \leq \frac{728}{541} 864^s \leq 1.346 n^{\log 864}.$$

Expanding $[\alpha, W_n]$ letter by letter, $U(n) \leq 2 \text{Rw}(n) + 5|W_n| \leq 2.70 n^{\log 864} + 5 n^{\log 323}$. For $m \geq 5$, $A(m) \leq 2U(m-2) + 18m - 35 \leq 5.39 m^{9.755} + 10 m^{8.336} + 18m \leq 0.87 m^{11}$, the ratio being largest at $m = 5$; and $A(2), A(3), A(4) \leq 1, 21, 47$. $\square$

Proof of Theorem 3.2. The proof above bounds every lifted cell by the largest area 864 of the images $\psi(r_i)$. Count instead the cells of each type. Let C_{ij} be the number of cells of type r_j in the certified filling of $\psi(r_i)$ from Lemma A.2; the certificates give

$$C = \begin{pmatrix} 4 & 0 & 1 & 8 & 4 \\ 108 & 32 & 31 & 128 & 60 \\ 204 & 68 & 64 & 384 & 144 \\ 0 & 0 & 0 & 0 & 0 \\ 36 & 12 & 14 & 84 & 36 \end{pmatrix},$$

with row sums 17, 359, 864, 0, 182. For $w = (1, \frac{63}{2}, \frac{338}{5}, 0, \frac{29}{2})$, exact arithmetic gives $Cw \leq 130w$ and $C\mathbf{1} \leq 17w$ entrywise. The weight of r_4 is zero, but $\psi(r_4)$ is freely trivial, so cells of type r_4 disappear under ψ ; the second inequality pays for the ordinary cells created by the last application of ψ . Thus if a diagram has weighted count S (a cell of type r_j counted with

weight w_j), substituting the certified fillings gives weighted count at most $130S$ and ordinary area at most $17S$. The relations $\rho^b = R'$ and $\psi(\rho) = R$ have ordinary costs 14 and 84 and weighted costs 48 and 288.

Let T_n and R_n count the letters $t^{\pm 1}$ and $\rho^{\pm 1}$ of W_n . The even step multiplies the column vector (T, R) by $D_0 = \begin{pmatrix} 2 & 26 \\ 1 & 20 \end{pmatrix}$, since $t \mapsto t\rho t$ and R has 26 letters $t^{\pm 1}$ and 20 letters $\rho^{\pm 1}$, and conjugation by b multiplies it by $B_0 = \begin{pmatrix} 1 & 4 \\ 1 & 3 \end{pmatrix}$. Since $(1, 18)B_0D_0 = (96, 1654) \leq 96(1, 18)$ and $(1, 18)D_0 \leq 96(1, 18)$, the quantity $L_n = T_n + 18R_n$ satisfies $L_n \leq 96L_{\lfloor n/2 \rfloor} + 1$, so $L_n \leq \frac{96}{95}96^s$ with $s = \lfloor \log n \rfloor$.

Let S_n be the weighted count of the recursive filling of $X_nW_n^{-1}$ and $m = \lfloor n/2 \rfloor$. Both branches give $S_n \leq 130S_m + 48T_m + 1248R_m \leq 130S_m + \frac{208}{3}L_m$, hence $S_n \leq \frac{3328}{1615}(130^s - 96^s) \leq \frac{21}{10}130^s$. For the ordinary count, the second inequality applied to the diagram of level m and the ordinary cost $14T_m + 364R_m$ of the new relations give $\text{Rw}(n) \leq 17S_m + \frac{182}{9}L_m \leq \frac{9}{20}130^s$, because $17 \cdot \frac{21}{10} + \frac{182}{9} \cdot \frac{96}{95} < \frac{9}{20} \cdot 130$. Expanding $[\alpha, W_n]$ letter by letter, $U(n) \leq 2\text{Rw}(n) + T_n + 5R_n \leq (\frac{9}{10} + \frac{96}{19})n^{\log 130}$. Finally $A(m) \leq 2U(m-2) + 18m - 35$ for $m \geq 3$ gives $A^*(M) \leq [2(\frac{9}{10} + \frac{96}{19}) + 18]M^{\log 130} = \frac{2841}{95}M^{\log 130} < 30M^{\log 130}$, and $A(2) = 1$. $\square$

B From far commutators to the full Dehn function

Proof of Theorem 3.3. Fix an integer $n \geq 1$ and put $F_n = 1 + A^*(4n + 2)$ and $U(0) = 0$. Lemma A.1 gives

$$U(q) \leq \sum_{i=0}^{q-1} E(i) = O(n^2 F_n) \quad (0 \leq q \leq 2n).$$

The implied constants in this proof are absolute.

Buffered collection. Let w be a null word of length at most n and put $h = a^n b^n$. The two quotient coordinates while reading w after h stay in $[0, 2n]^2$. Write the current word as $v s(m, q)$, where $s(m, q) = a^m b^q$ and v is a word in the $\sigma_j^{\pm 1}$. Using $bab^{-1} = \alpha^{-1}a$ exactly q times gives $b^q ab^{-q} = \tau_{q-1}^{-1} \cdots \tau_0^{-1} a$. Replace each τ_j by σ_j at cost $U(j)$. Thus

$$\begin{aligned} s(m, q)a &= (\sigma_{m+q-1}^{-1} \cdots \sigma_m^{-1})s(m+1, q), \\ s(m, q)a^{-1} &= (\sigma_{m-1} \cdots \sigma_{m+q-2})s(m-1, q) \end{aligned}$$

each costs at most $q + \sum_{j < q} U(j) = O(n^3 F_n)$; an empty product is omitted. The second rule is the inverse rearrangement of the first at $(m-1, q)$, and is used only when the next coordinate is nonnegative. The rules $s(m, q)\alpha^{\pm 1} = \sigma_{m+q}^{\pm 1}s(m, q)$ cost $U(q)$, while multiplication by $b^{\pm 1}$ merely changes q and costs no cells. The buffer ensures that all indices produced are nonnegative.

After at most n rules the quotient coordinates return to (n, n) , and hence hwh^{-1} has been changed to a word v at cost $O(n^4 F_n)$. It has at most $2n^2 + n$ letters, with $0 \leq j \leq 4n$. Since $w = 1$, the word v is the identity permutation on the first $4n + 2$ points of ray 1. It remains to fill this permutation word with a cost that is uniform in the size of that symmetric group.

Coxeter insertion. In S_{d+1} , with adjacent transpositions $s_0, \dots, s_{d-1}$, use the recursive normal form uc_k , where $u \in S_d$, $c_k = s_{d-1}s_{d-2} \cdots s_k$ for $0 \leq k < d$, and $c_d = 1$. With right actions c_k sends the last point to $k+1$; this determines k , after which u is determined recursively. To insert a generator on the right use

$$c_k s_i = \begin{cases} s_i c_k, & i \leq k-2, \\ c_{k-1}, & i = k-1, \\ c_{k+1}, & i = k, \\ s_{i-1} c_k, & i \geq k+1. \end{cases}$$

The first case uses far commutations, the middle cases concatenate or cancel, and the last case moves the final s_i left across $s_k, \dots, s_{i-2}$, applies one braid, and moves the resulting s_{i-1} left across $s_{i+1}, \dots, s_{d-1}$. There are at most $d + 1$ Coxeter moves at a level and at most d levels per insertion, since only one generator is passed into u . Consequently a word of length l reduces to its unique normal form in $O(ld^2)$ moves, and the identity reduces to the empty word.

Apply this with $d = 4n + 1$ to v . First replace inverse letters by positive ones using $\sigma_j^2 = 1$, at one cell per letter. A Coxeter square costs one cell in P , a braid costs at most four, and a far commutation costs at most $A^*(4n + 2)$. Thus the $O(n^2)$ letters of v fill at cost $O(n^4 F_n)$. Combining with collection and using invariance of area under free conjugation proves the asserted bound for $\delta_P(n)$. Substituting Theorem 3.2, or Theorem 3.1, gives the two polynomial exponents. $\square$

Use of AI assistance

Generative AI tools assisted with developing arguments and constructions, checking proofs and numerical identities, searching the literature, writing verification code, and drafting and editing the manuscript. Automated review is not a substitute for mathematical proof. The accompanying verification notes distinguish executable checks and kernel-checked Lean results from proofs in the manuscript that are not formalized in Lean. The authors take responsibility for the content.

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